

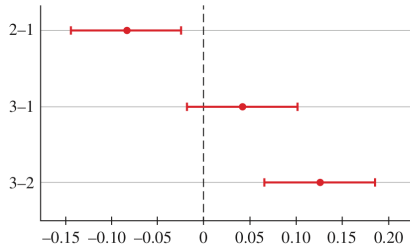
14. Calcium, again Refer to Exercise 13 (Section 11.2), where the calcium content of a powdered mineral substance was analyzed using three different methods. The comparisons option in *MINITAB* provided the printout that follows.

Tukey Simultaneous Test for Differences of Means

Difference of Levels	Difference of Means	SE of Difference	95% CI	T-Value	Adjusted P-Value
2-1	-0.0840	0.0224	(-0.1438, -0.0242)	-3.75	0.007
3-1	0.0420	0.0224	(-0.0178, 0.1018)	1.87	0.189
3-2	0.1260	0.0224	(0.0662, 0.1858)	5.62	0.000

Individual confidence level = 97.94%

Tukey Simultaneous 95% CIs Differences of Means for Calcium



If an interval does not contain zero, the corresponding means are significantly different.

Use Tukey's method to compare the three population means. How would you summarize your results?

- If the interval contains 0 $\Rightarrow$ not significantly different
- If the interval does not contain 0 $\Rightarrow$ significantly different

2-1 : $(-0.1438, -0.0242) \rightarrow$ does not contain 0

$\therefore$ significantly different

3-1 : $(-0.0178, 0.1018) \rightarrow$ contains 0

$\therefore$ not significantly different

3-2 : $(0.0662, 0.1858) \rightarrow$ does not contain 0

$\therefore$ significantly different

$\therefore$ populations 1 & 3 are not significantly different, population 2 is significantly different lower than both.

16. The Cost of Lumber, continued The analysis of variance F -test in Exercise 15 (Section 11.2) indicated that there was a difference in the average cost of lumber for the four states. The following information from that exercise is given in the table:

Sample Means	$\bar{x}_1 = 262.2$	MSE	41.25
	$\bar{x}_2 = 234.8$	Error df:	16
	$\bar{x}_3 = 251.6$	n_j :	5
	$\bar{x}_4 = 268.6$	k :	4

Use Tukey's method for paired comparisons to determine which means differ significantly from the others at the $\alpha = .01$ level.

$$\alpha = 0.01$$

$$k = 4$$

$$df_E = 16$$

$$q_{0.01(4, 16)} = 5.19$$

$$\begin{aligned} \text{HSD} &= q \sqrt{\frac{\text{MSE}}{n}} \\ &= 5.19 \sqrt{\frac{41.25}{5}} \\ &= 14.91 \end{aligned}$$

$$|\bar{x}_1 - \bar{x}_2| = 27.4 > 14.91, \text{ significant difference}$$

$$|\bar{x}_1 - \bar{x}_3| = 10.6 < 14.91, \text{ not significant difference}$$

$$|\bar{x}_1 - \bar{x}_4| = 6.4 < 14.91, \text{ not significant difference}$$

$$|\bar{x}_2 - \bar{x}_3| = 16.8 > 14.91, \text{ significant difference}$$

$$|\bar{x}_2 - \bar{x}_4| = 33.8 > 14.91, \text{ significant difference}$$

$$|\bar{x}_3 - \bar{x}_4| = 17.0 > 14.91, \text{ significant difference}$$

$\therefore$ At $\alpha = 0.01$, mean 2 is significantly different from means 1, 3, 4. Mean 3 is significantly different from mean 4. Mean 1 is not significantly different from mean 3 & 4.

17. GRE Scores The quantitative reasoning scores on the Graduate Record Examination (GRE)³ were recorded for students admitted to three different graduate programs at a local university.

Graduate Program					
Life Sciences		Physical Sciences		Social Sciences	
630	660	660	760	440	540
640	660	640	670	330	450
470	480	720	700	670	570
600	650	690	710	570	530
580	710	530	450	590	630
$\bar{x}_1 = 608$		$\bar{x}_2 = 653$		$\bar{x}_3 = 552$	

- Do these data provide sufficient evidence to indicate a difference in the mean GRE scores for applicants admitted to the three programs?
- Find a 95% confidence interval for the difference in mean GRE scores for Life Sciences and Physical Sciences.
- If you find a significant difference in the average GRE scores for the three programs, use Tukey's

method for paired comparisons to determine which means differ significantly from the others. Use $\alpha = .05$.

$$a) CM = \frac{(\sum \text{total})^2}{N} = \frac{17930^2}{30} = 10716163.33$$

$$TSS = \sum x^2 - CM = 11014900 - 10716163.33 = 306736.67$$

$$SST = \frac{\sum x_i^2}{n} - CM = \frac{6080^2 + 6530^2 + 5320^2}{10} - 10716163.33 = 74806.67$$

$$SSE = TSS - SST = 225930$$

$$MST = \frac{SST}{k-1} = \frac{74806.67}{2} = 37403.33$$

$$MSE = \frac{SSE}{N-k} = \frac{225930}{27} = 8367.78$$

$$F = \frac{MST}{MSE} = 4.47$$

$$H_0: \mu_1 = \mu_2 = \mu_3$$

$$H_a: \text{not all } \mu_i \text{ are equal}$$

$$df_1 = 2, df_2 = 27, \alpha = 0.05 \rightarrow 3.35$$

$$\therefore 4.47 > 3.35 \rightarrow \text{RR}$$

$\therefore$ have sufficient evidence that the mean GRE scores are different for the three programs.

$$b) (\bar{x}_1 - \bar{x}_2) \pm t_{0.025, 27} \cdot SE$$

$$= (608 - 653) \pm 2.052 \times \sqrt{8367.78 \left(\frac{2}{10}\right)}$$

$$= -45 \pm 2.052 (40.91)$$

$$= (-128.94, 38.94)$$

$$c) HSD = q_{0.05, 3, 27} \sqrt{\frac{MSE}{n}}$$

$$= 3.506 \sqrt{\frac{8367.78}{10}}$$

$$= 101.43$$

$$|\bar{x}_1 - \bar{x}_2| = 45 < 101.43$$

$$|\bar{x}_1 - \bar{x}_3| = 76 < 101.43$$

$$|\bar{x}_2 - \bar{x}_3| = 121 > 101.43$$

} not significantly different

$\rightarrow$ significantly different

24. Where to Shop? Do you shop at the grocery store closest to home or do you look for the store that has the best prices? We compared the regular prices at four different grocery stores for eight items purchased on the same day.

Items	Stores			
	Vons	Ralphs	Stater Bros	WinCo
Salad mix, 360 g bag	3.99	2.79	1.99	1.78
Hillshire Farm® Beef Smoked Sausage, 420 g	4.29	4.29	3.99	2.50
Kellogg's Raisin Bran®, 750 g	4.49	5.49	4.49	3.15
Kraft® Philadelphia® Cream Cheese, 250 g	2.99	3.19	2.79	1.48
Kraft® Ranch Dressing, 450 mL	3.19	3.49	3.49	1.48
Treetop® Apple Juice, 2 L	2.99	3.49	3.49	1.58
Dial® Bar Soap, Gold, 125 g	5.99	6.49	5.79	5.14
Jif® Peanut Butter, Creamy, 850 g	5.15	5.49	4.79	4.34

- What are the blocks and treatments in this experiment?
- Do the data provide evidence to indicate that there are significant differences in prices from store to store? Support your answer statistically using the ANOVA printout that follows.
- Are there significant differences from block to block? Was blocking effective?

ANOVA: Price versus Item, Store

Analysis of Variance for Price

Source	DF	SS	MS	F	P
Store	3	13.192	4.3974	26.52	0.000
Item	7	40.995	5.8564	35.32	0.000
Error	21	3.482	0.1658		
Total	31	57.669			

Model Summary

S	R-sq	R-sq(adj)
0.407181	93.96%	91.09%

a) block : 8 items
treatments : 4 stores

b) $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$
 H_a : Not all store means are equal
 From ANOVA table for store:
 $F = 26.52$, $p = 0.000$

$\therefore p < 0.05$

$\therefore$ Reject H_0 , there is sufficient evidence that mean prices differ from store to store.

c) From ANOVA table for item row :
 $F = 35.32$, $p = 0.000$

$\therefore p < 0.05$

$\therefore$ reject H_0 : all block means are equal. Yes, there are significant differences from block to block. Blocking is effective.

25. Where to Shop?, continued Refer to Exercise 24.

The printout that follows provides the average costs of the selected items for the $k = 4$ stores.

Means

Store	N	Price
Ralphs	8	4.34000
Stater Bros	8	3.85250
Vons	8	4.13500
WinCo	8	2.68125

- What is the appropriate value of $q_{\alpha}(k, df)$ for testing for differences among stores?
- What is the value of $\omega = q_{\alpha}(k, df) \sqrt{\frac{MSE}{b}}$?
- Use Tukey's pairwise comparison test among stores used to determine which stores differ significantly in average prices of the selected items.

$$a) q_{0.5}(k, df_E) = q_{0.5}(4, 21) \\ = 3.96$$

$$b) W = q_{0.5}(k, df) \sqrt{\frac{MSE}{b}} \\ = 3.96 \sqrt{\frac{0.1658}{8}} \\ = 0.57$$

$$c) \left. \begin{array}{l} |\bar{x}_1 - \bar{x}_2| = 0.4875 < W \\ |\bar{x}_1 - \bar{x}_3| = 0.205 < W \\ |\bar{x}_1 - \bar{x}_4| = 1.658 > W \rightarrow \text{significantly different} \\ |\bar{x}_2 - \bar{x}_3| = 0.28 < W \rightarrow \text{not significantly different} \\ |\bar{x}_2 - \bar{x}_4| = 1.171 > W \\ |\bar{x}_3 - \bar{x}_4| = 1.45 > W \end{array} \right\} \text{significantly different}$$

$\therefore$ WinCo is significantly lower than each of the other three stores.

26. Rocket Propellants An experiment was conducted to compare three mixtures of components in making rocket propellant. Five samples of each mixture were obtained for testing. Five investigators measured the propellant thrust in pounds for each of the mixtures, with the mixtures presented in random order.

Investigators	Mixtures		
	1	2	3
1	42	39	45
2	37	36	40
3	43	48	50
4	41	47	52
5	39	42	47

$T_1 = 209$ $T_2 = 210$ $T_3 = 239$

- Identify the treatments and the blocks in this experiment and provide an appropriate analysis of variance.
- Is there sufficient evidence to conclude there is a significant difference in the propellant thrust for the mixtures? Use $\alpha = .01$.
- Was blocking effective? Use the p -value to draw your conclusions and justify your answer.
- Use Tukey's pairwise comparisons with $\alpha = .01$ to determine which propellant mixtures differed from others. Summarize your results.

A) Treatments : Mixtures

Blocks : Investigators

$$b = 5, k = 3$$

$$CM = \frac{648^2}{15} = 27993.6$$

$$TSS = \sum_{i=1}^3 \sum_{j=1}^5 ij^2 - CM = 322.4$$

$$SST = \sum_{j=1}^3 \frac{(X_j)^2}{k} = \frac{84510}{3} - CM = 176.4$$

$$SSE = 322.4 - 107.2 - 176.4$$

Source	df	SS	MS	F
Mixtures	2	107.2	53.6	11.05
Investigators	4	176.4	44.1	9.09
Error	8	38.8	4.85	
Total	14	322.4		

b) $H_0: \mu_1 = \mu_2 = \mu_3$

H_a : not all mixture means are equal

$$F_{2,8,0.01} = 8.65$$

$$F = 11.05 > 8.65$$

$\therefore$ reject H_0 , have sufficient different

c) $F = 9.09$, $p\text{-value} = 0.0045$

$$\therefore p < 0.01$$

$$d) W = q \times \sqrt{\frac{MSE}{b}} = 6.96 \times \sqrt{\frac{4.85}{5}} = 6.85$$

$$|\bar{x}_2 - \bar{x}_1| = 6.4 < W$$

$$|\bar{x}_3 - \bar{x}_2| = 4.4 < W$$

$$|\bar{x}_1 - \bar{x}_3| = 2 < W$$

} not significantly different